

# 2026-07-11 - Handout – Sliding Window

## Q1. Longest Substring Without Repeating Characters [MED]

Link: <https://leetcode.com/problems/longest-substring-without-repeating-characters/>

Given a string **s**, find the length of the **longest** substring without duplicate characters.

### Constraints:

$0 \leq s.length \leq 5 * 10^4$

s consists of English letters, digits, symbols and spaces.

### Example 1:

**Input:** s = "abcabcbb"

**Output:** 3

**Explanation:** The answer is "abc", with the length of 3. Note that "bca" and "cab" are also correct answers.

### Example 2:

**Input:** s = "bbbbbb"

**Output:** 1

**Explanation:** The answer is "b", with the length of 1.

## Q2. Fruit Into Basket [MED]

Link: <https://leetcode.com/problems/fruit-into-baskets/>

You are visiting a farm that has a single row of fruit trees arranged from left to right. The trees are represented by an integer array **fruits** where **fruits[i]** is the **type** of fruit the **i**th tree produces.

You want to collect as much fruit as possible. However, the owner has some strict rules that you must follow:

- You only have **two** baskets, and each basket can only hold a **single type** of fruit. There is no limit on the amount of fruit each basket can hold.
- Starting from any tree of your choice, you must pick **exactly one fruit** from **every** tree (including the start tree) while moving to the right. The picked fruits must fit in one of your baskets.
- Once you reach a tree with fruit that cannot fit in your baskets, you must stop.

Given the integer array **fruits**, return the **maximum** number of fruits you can pick.

### Constraints:

$1 \leq fruits.length \leq 10^5$

$0 \leq fruits[i] < fruits.length$

### Example 1:

**Input:** fruits = [1,2,1]

**Output:** 3

**Explanation:** We can pick from all 3 trees.

### Example 2:

**Input:** fruits = [0,1,2,2]

**Output:** 3

**Explanation:** We can pick from trees [1,2,2].

If we had started at the first tree, we would only pick from trees [0,1].

### Q3. Minimum Window Substring [HARD]

Link: <https://leetcode.com/problems/minimum-window-substring/>

Given two strings **s** and **t** of lengths **m** and **n** respectively, *return the minimum window substring of s such that every character in t (including duplicates) is included in the window*. If there is no such substring, return the empty string "".

The testcases will be generated such that the answer is **unique**.

**Constraints:**

$m == s.length$

$n == t.length$

$1 \leq m, n \leq 105$

s and t consist of uppercase and lowercase English letters.

**Example 1:**

**Input:** s = "ADOBECODEBANC", t = "ABC"

**Output:** "BANC"

**Explanation:** The minimum window substring "BANC" includes 'A', 'B', and 'C' from string t.

**Example 2:**

**Input:** s = "a", t = "a"

**Output:** "a"

**Explanation:** The entire string s is the minimum window.

### Q4. Permutation in String [MED]

Link: <https://leetcode.com/problems/permutation-in-string/>

Given two strings **s1** and **s2**, return **true** if **s2** contains a permutation of **s1**, or **false** otherwise.

In other words, return **true** if one of **s1**'s permutations is the substring of **s2**.

**Constraints:**

$1 \leq s1.length, s2.length \leq 104$

s1 and s2 consist of lowercase English letters.

**Example 1:**

**Input:** s1 = "ab", s2 = "eidbaooo"

**Output:** true

**Explanation:** s2 contains one permutation of s1 ("ba").

**Example 2:**

**Input:** s1 = "ab", s2 = "eidboaoo"

**Output:** false